1.7 Basic Matrix Operations

Name _____ Period ____ Date _____

Algebra II 1.7 · Student sheet

I can…

- add and subtract matrices.
- multiply a matrix by a scalar and evaluate matrix expressions.
- solve matrix equations.

Vocabulary · fill in the missing word

Equal matrices — same dimensions and equal _____ entries

Matrix addition and subtraction — add or subtract _____ entries; the matrices must have the same dimensions

Scalar multiplication — multiply every _____ by the number (the scalar)

Zero matrix — every _____ is 0

Part 1 · Store inventory

Rows: Store A, Store B. Columns: shirts, hats, mugs.

Week 1: W1 = [12 8 5; 10 6 9] Week 2: W2 = [15 7 4; 9 10 6]

W1 + W2 (total)	W2 − W1 (change)	2 · W1 (double)

Part 2 · Notes: fill in the blanks

Matrices can be added only when they have the same _____.

To add, combine _____ entries.

Scalar multiplication: multiply _____ entry by the scalar.

Part 3 · Examples (work them with the class)

Example 1

A = [3 −1; 0 4] and B = [−2 5; 6 −3]. Find A + B and A − B.

Example 2

Find −3 · [2 0; −1 4].

Example 3

Use A and B from Example 1 to find 2A − B.

Example 4

Solve for x, y, and z: [x 4; −2 y] + [3 1; z 2] = [10 5; 0 −3]

Example 5

Find the matrix X: X + [1 2; 3 4] = [5 0; −1 6]

Part 4 · Partner practice

Partner 1

Find [4 −2; 1 0] − [−1 3; 5 2].

Partner 2

Find X: 2X − [2 0; 4 −6] = [8 2; −2 0]